

Student Management System (C++ Project Code)

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#include <iostream>
#include <fstream>
#include <vector>
using namespace std;

class Student {
public:
    int id;
    string name;
    int age;

    void input() {
        cout << "Enter ID: ";
        cin >> id;
        cout << "Enter Name: ";
        cin >> name;
        cout << "Enter Age: ";
        cin >> age;
    }

    void display() {
        cout << "ID: " << id << " | Name: " << name << " | Age: " << age << endl;
    }
};

vector<Student> students;

void saveToFile() {
    ofstream file("students.txt");
    for (auto s : students) {
        file << s.id << " " << s.name << " " << s.age << endl;
    }
    file.close();
}

void loadFromFile() {
    ifstream file("students.txt");
    Student s;
    while (file >> s.id >> s.name >> s.age) {
        students.push_back(s);
    }
    file.close();
}

void addStudent() {
    Student s;
    s.input();
    students.push_back(s);
    saveToFile();
}

void displayStudents() {
    for (auto s : students) {
        s.display();
    }
}

void searchStudent() {
    int id;
    cout << "Enter ID to search: ";
    cin >> id;

    for (auto s : students) {
        if (s.id == id) {
            s.display();
            return;
        }
    }
    cout << "Student not found!" << endl;
}

void deleteStudent() {
    int id;
```

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        cout << "Enter ID to delete: ";
        cin >> id;

        for (int i = 0; i < students.size(); i++) {
            if (students[i].id == id) {
                students.erase(students.begin() + i);
                saveToFile();
                cout << "Deleted successfully!" << endl;
                return;
            }
        }
        cout << "Student not found!" << endl;
    }

int main() {
    loadFromFile();
    int choice;

    do {
        cout << "\n--- Student Management System ---\n";
        cout << "1. Add Student\n";
        cout << "2. Display Students\n";
        cout << "3. Search Student\n";
        cout << "4. Delete Student\n";
        cout << "5. Exit\n";
        cout << "Enter choice: ";
        cin >> choice;

        switch (choice) {
            case 1: addStudent(); break;
            case 2: displayStudents(); break;
            case 3: searchStudent(); break;
            case 4: deleteStudent(); break;
            case 5: cout << "Exiting...\n"; break;
            default: cout << "Invalid choice!\n";
        }

    } while (choice != 5);

    return 0;
}

```